

ENG // UTILITY CONSOLE

UI/UX Final Review · 2026-07-12 · huseyinozdes/randomwhattheflyingduck

PRODUCT OVERVIEW

A single-page static HTML engineering utility console served directly in the browser. Five functional tabs: **IMAGE** (load / bg-remove / export raster), **DATA** (structured CSV editor), **CONVERSION** (bidirectional unit conversions), **CAD** (OBJ→STL / STL pass-through), and **DATE/TIME** (date arithmetic + clipboard copy). All processing runs client-side; no data leaves the session.

ACCEPTABILITY VERDICT

✓ **PRODUCTION-ACCEPTABLE** — All critical and major issues have been resolved in this pass. Remaining items are optional quality improvements.

ISSUES FIXED IN THIS PASS (PR #9)

Sev	Issue	Resolution
CRITICAL	Tab bar at the bottom of <main> Users couldn't see navigation without scrolling	Moved sheet-tab-bar to top of <main>; flipped CSS border/active-accent direction
CRITICAL	No ARIA roles on tab component Keyboard/AT navigation broken	Added role=tablist/tab/tabpanel, aria-selected, aria-controls, Arrow/Home/End keyboard nav
MAJOR	copyOutput() silently called wrong status bar — no feedback on DATE/TIME tab	Added #copy-toast fixed element + showToast() helper
MAJOR	swapConversion() didn't recalculate Fields out of sync after swap	Calls convert function after each swap
MEDIUM	Duplicate input+change listener on file input	Removed redundant input listener
MEDIUM	Date inputs render light-mode browser chrome on dark UI	Added color-scheme: dark to date/number inputs

OPTIONAL TODO BACKLOG — NO RELEASE BLOCKERS

The following items are improvements worth tracking but are not required for production use. Implement in priority order when bandwidth allows.

Priority	TODO Item	Detail
OPT - 1	Drag-and-drop file loading (IMAGE + CAD tabs)	Add dragover/drop handlers on the preview/drop zones. Use DataTransfer.files[0] → existing loadSelectedImage() / cadFileInput flow. Improves power-user speed.
OPT - 2	Filename truncation tooltip	Add title='<full filename>' to #file-name and #cad-file-name elements so full names are accessible on hover.

Priority	TODO Item	Detail
OPT - 3	Conversion decimal precision normalization	in→mm uses 3 dp; mm→in uses 4 dp; ft→m uses 4 dp. Standardise to 4 dp across all pairs for consistency.
OPT - 4	fallbackCopy uses deprecated document.execCommand	execCommand('copy') is deprecated. Replace fallback with a graceful 'clipboard unavailable' toast message (the Clipboard API already handles the happy path).
OPT - 5	Image preview fixed height 220px	For portrait-heavy images the 220px cap wastes space; for landscape it may be tall. Consider a max-height with auto-height that collapses when no image is loaded.
OPT - 6	Data tab: no column-header row validation	The textarea accepts any text. Consider a simple first-row header check before CSV export to warn if no header row is present.
OPT - 7	CAD tab: no accept filter on file input	Add accept='.obj,.stl,.dxf,.dwg,.step,.stp,.iges,.igs' to #cad-file-input to filter the OS file picker.

Review scope: static analysis of index.html + mini_tool_engine_adapter.js. Branch: huseyinozdes-duck-uiux-production-pass · PR #9 merged into main. Generated 2026-07-12 by Copilot CLI.